

ELECTRIC VEHICLE TECHNOLOGY INTERNSHIP

TASK 3: EV CHARGING ANALYSIS

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Objective

To study major EV charging levels and technologies, compare their charging power, time and applications, and estimate the charging time of a selected EV battery pack.

Selected Battery Pack for Calculation

96S2P lithium-ion battery configuration: 355.2 V nominal pack voltage, 100 Ah capacity and approximately 35.52 kWh nominal energy.

1. EV Charging Levels and Technologies

EV charging can be broadly grouped into AC charging and DC fast charging. AC charging normally uses the vehicle's onboard charger, while DC fast charging supplies DC power to the battery through suitable high-power charging equipment.

Charging type	Typical power	Typical use	Key characteristics
AC Level 1	≈1.4–1.9 kW	Home / overnight	Lowest power; long charging time
AC Level 2	≈7–19.2 kW	Home, workplace, fleet	Faster AC charging; common for regular daily charging
DC Fast Charging	≈50–150+ kW	Highway / public fast charging	High charging power; charging rate depends on vehicle
High-power DC	≈150–400 kW	Highway / rapid charging	Very fast charging; requires compatible vehicle and charger

Charging connectors and technologies

Common conductive charging systems include SAE J1772 for AC charging in North America and CCS for DC fast charging. CHAdeMO and NACS are also used in different markets and vehicle ecosystems. Wireless charging is another technology covered by SAE J2954, designed around interoperability, safety and performance requirements.

Source basis: U.S. Department of Energy EV charging guidance and SAE charging standards. Power ranges vary by installation, vehicle and charger.

2. Comparison of Charging Options

Parameter	AC Level 1	AC Level 2	DC Fast
Current type	AC	AC	DC
Representative power	1.4–1.9 kW	7–19.2 kW	50–150+ kW
Typical location	Home	Home / workplace	Public / highway
Charging speed	Slow	Moderate	Fast
Infrastructure cost	Low	Moderate	High
Best application	Overnight charging	Daily charging	Long-distance / quick top-up
Main limitation	Long charging time	Needs higher-power supply	Higher cost and battery thermal stress

Application summary

For routine overnight charging, low-power AC charging can be sufficient. AC Level 2 is more practical when faster turnaround is required at homes, workplaces or fleets. DC fast charging is suited to public stations and long-distance travel because it can deliver substantially higher power, although actual charging speed is limited by the vehicle's maximum charging rate and the battery's state of charge and temperature.

Key takeaway: The fastest charger is not always the best charger. The appropriate option depends on required turnaround time, available electrical infrastructure, vehicle charging capability, battery condition and cost.

3. Charging Time Calculation for Selected Battery Pack

Selected pack: 96S2P | Nominal voltage = 355.2 V | Capacity = 100 Ah | Nominal energy = 35.52 kWh.

Basic formula

Charging time (hours) $\approx$ Battery energy (kWh) $\div$ [Charger power (kW) $\times$ charging efficiency]

For a simple estimate, 90% overall charging efficiency is assumed. Real-world charging time can be longer because charging power may reduce near high state of charge and because of thermal and vehicle-side limits.

Charger	Power	Ideal time	Approx. time @ 90%
AC Level 1 (1.5 kW)	1.5 kW	23 h 41 min	26 h 19 min
AC Level 2 (7.2 kW)	7.2 kW	4 h 56 min	5 h 29 min
AC Level 2 (11 kW)	11.0 kW	3 h 14 min	3 h 35 min
DC Fast (50 kW)	50.0 kW	0 h 43 min	0 h 47 min
DC Fast (100 kW)	100.0 kW	0 h 21 min	0 h 24 min
High-power DC (150 kW)	150.0 kW	0 h 14 min	0 h 16 min

Worked example: 50 kW DC fast charging

Ideal time = $35.52 \div 50 = 0.7104$ h $\approx$ 42.6 minutes. With 90% charging efficiency, estimated time = $35.52 \div (50 \times 0.90) \approx 0.789$ h $\approx$ 47 minutes. Actual 0–100% time can be higher because DC charging typically tapers at high state of charge.

4. Findings and Conclusion

- AC Level 1 provides the lowest charging power and is mainly suitable for overnight or low-mileage use.
- AC Level 2 provides a practical balance between charging speed, installation requirements and everyday usability.
- DC fast charging greatly reduces charging time and is useful for highway travel, fleets and quick top-ups.
- For the selected 35.52 kWh battery pack, theoretical charging time decreases substantially as charger power increases, but real charging is controlled by the vehicle, battery temperature, state of charge and charger limits.
- Charging infrastructure should be selected according to the vehicle's maximum accepted power rather than charger rating alone.

Conclusion

EV charging is a key part of the electric mobility ecosystem. Understanding charging levels, power requirements and charging-time calculations helps engineers select suitable charging infrastructure. The analysis of the selected 35.52 kWh battery pack shows the trade-off between charging speed, power demand and practical charging conditions.

References

1. U.S. Department of Energy – EV charging levels and EVSE guidance.
2. U.S. Department of Energy Alternative Fuels Data Center – Electric Vehicle Charging Options.
3. SAE International – J1772 conductive charging and J2954 wireless power transfer standards.